

```
In [5]: import numpy as np

array = np.array([27, 50, 44, 6, 58, 61, 23, 86, 67, 20, 75, 7, 79, 61, 90, 54])

print(array)
```

```
[27 50 44  6 58 61 23 86 67 20 75  7 79 61 90 54]
```

```
In [6]: array.mean()
```

```
Out[6]: np.float64(50.5)
```

```
In [8]: np.percentile(array, 25)
```

```
Out[8]: np.float64(26.0)
```

```
In [9]: np.percentile(array, 50)
```

```
Out[9]: np.float64(56.0)
```

```
In [10]: np.percentile(array, 75)
```

```
Out[10]: np.float64(69.0)
```

```
In [11]: np.percentile(array, 100)
```

```
Out[11]: np.float64(90.0)
```

```
In [12]: def outDetection(array):
          sorted(array)
          Q1, Q3 = np.percentile(array, [25, 75])
          IQR = Q3 - Q1
          lr = Q1 - (1.5 * IQR)
          ur = Q3 + (1.5 * IQR)
          return lr, ur
```

```
In [13]: lr, ur = outDetection(array)
```

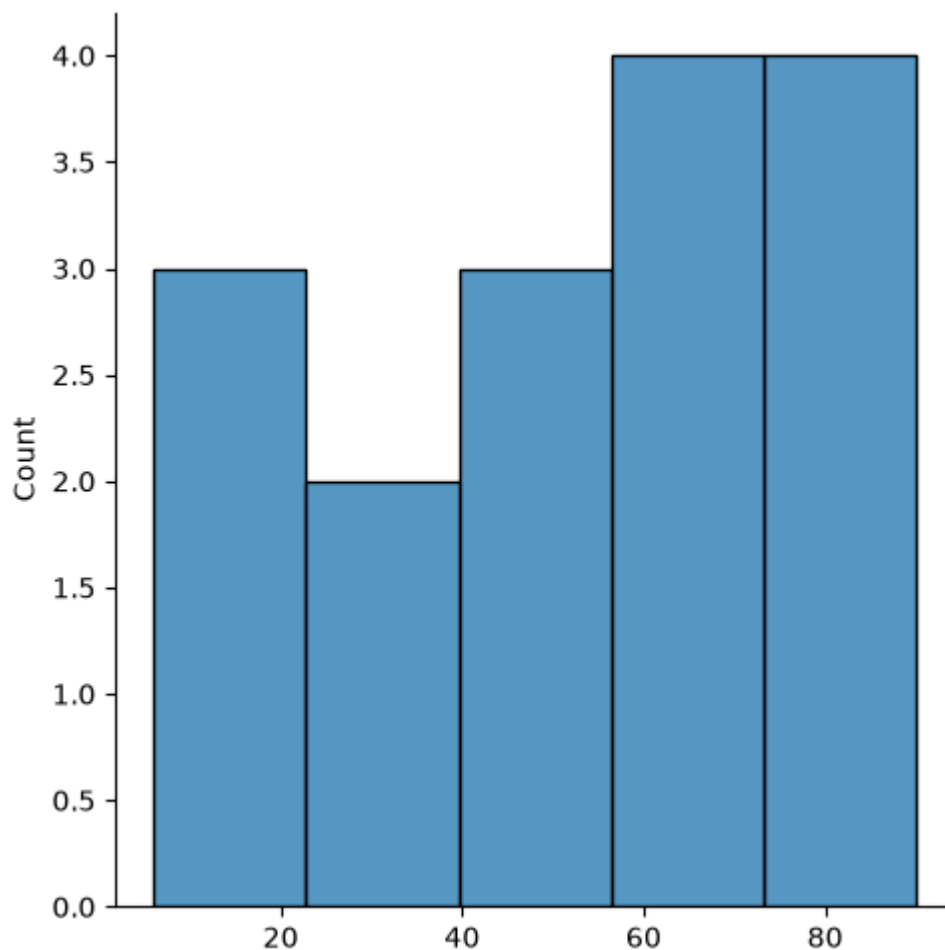
```
lr, ur
```

```
Out[13]: (np.float64(-38.5), np.float64(133.5))
```

```
In [14]: import seaborn as sns
          %matplotlib inline
```

```
In [15]: sns.displot(array)
```

```
Out[15]: <seaborn.axisgrid.FacetGrid at 0x1b798abd2b0>
```



```
In [16]: sns.distplot(array)
```

C:\Users\chaar\AppData\Local\Temp\ipykernel_32948\1855776059.py:1: UserWarning:

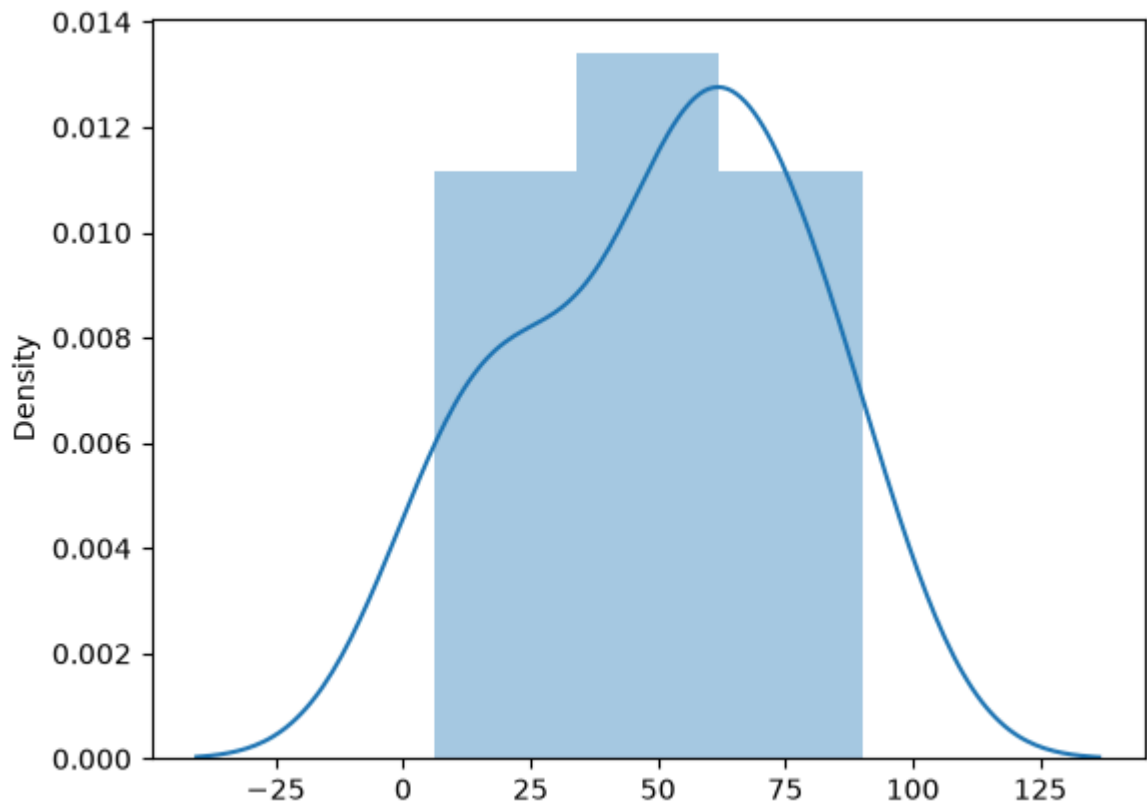
``distplot`` is a deprecated function and will be removed in seaborn v0.14.0.

Please adapt your code to use either ``displot`` (a figure-level function with similar flexibility) or ``histplot`` (an axes-level function for histograms).

For a guide to updating your code to use the new functions, please see <https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751>

```
sns.distplot(array)
```

```
Out[16]: <Axes: ylabel='Density'>
```



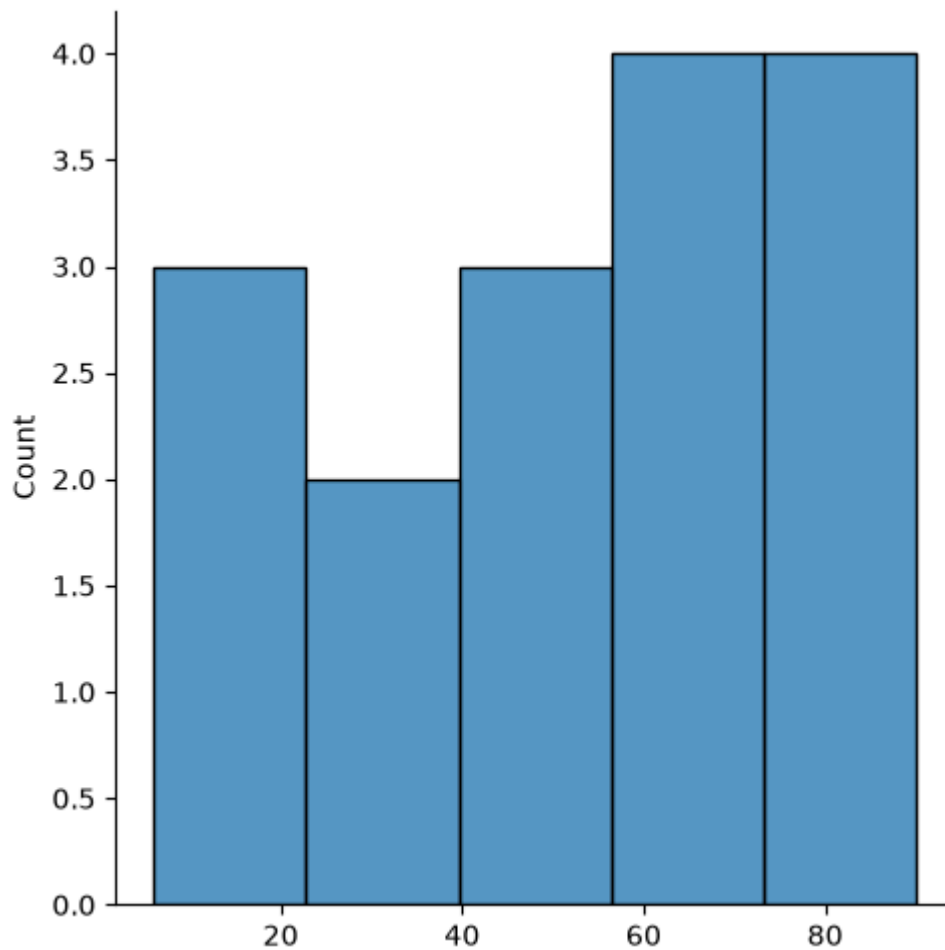
In []:

```
In [18]: new_array = array[(array > lr) & (array < ur)]  
  
new_array
```

```
Out[18]: array([27, 50, 44,  6, 58, 61, 23, 86, 67, 20, 75,  7, 79, 61, 90, 54])
```

```
In [19]: sns.displot(new_array)
```

```
Out[19]: <seaborn.axisgrid.FacetGrid at 0x1b79929e710>
```



```
In [20]: lr1, ur1 = outDetection(new_array)

lr1, ur1
```

```
Out[20]: (np.float64(-38.5), np.float64(133.5))
```

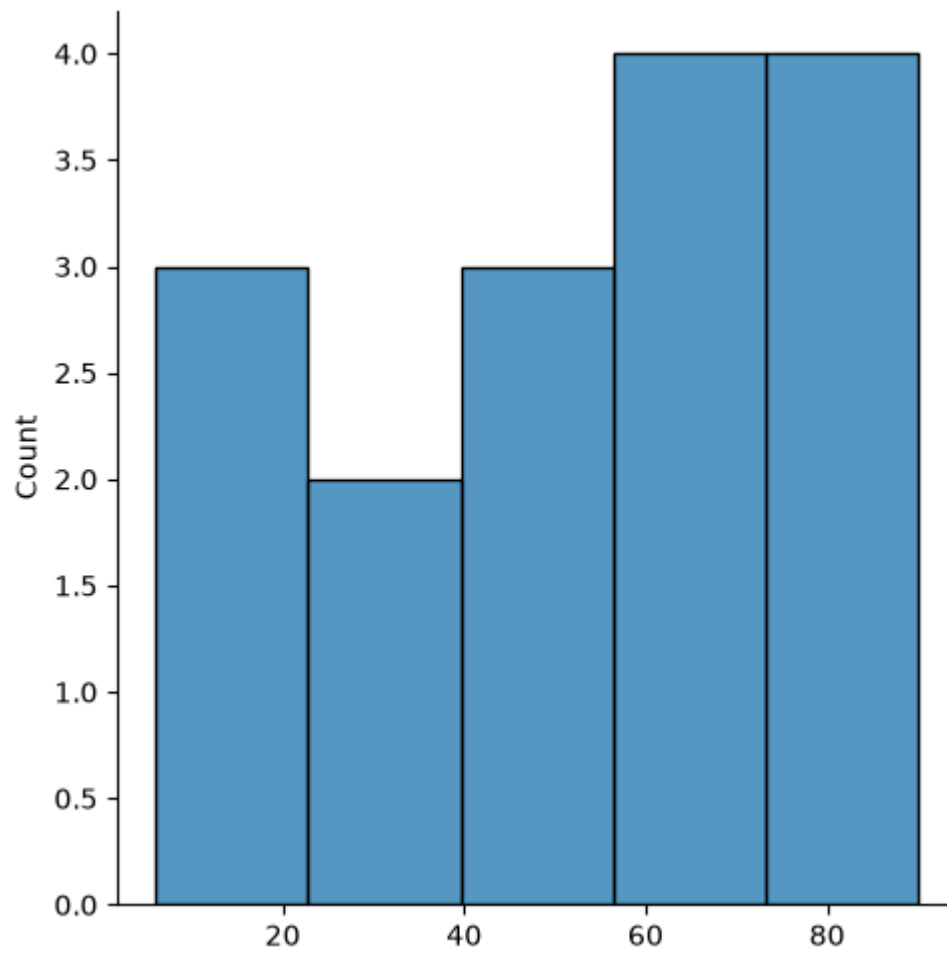
```
In [21]: final_array = new_array[(new_array > lr1) & (new_array < ur1)]

final_array
```

```
Out[21]: array([27, 50, 44,  6, 58, 61, 23, 86, 67, 20, 75,  7, 79, 61, 90, 54])
```

```
In [23]: sns.displot(final_array)
```

```
Out[23]: <seaborn.axisgrid.FacetGrid at 0x1b7993c6350>
```



In []: